Lacunary polynomials exclude the missing reverse Carleson measures

A candidate full answer to Problem 8.1 of arXiv:2412.02354

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Status: candidate full result, likely valid. Problem 8.1 has an affirmative answer. More strongly, for fine index below two, no finite-measure L^s norm can dominate either of the two mixed analytic spaces considered below, for any finite s.

1 Source problem

Doubtsov, Tselishchev, and Vasilyev study reverse Carleson inequalities for spaces of analytic functions in the disk [1]. On page 22, Problem 8.1 asks whether there are no reverse Carleson measures for $HB_0^{p,q}$ when $0 < p < 1$ and $p < q < 2$.

> Indeed, in Theorem 6.1 we have not considered the case $p < 1$. However, if $p < 1$ and $q \geq 2$ one can prove that a measure μ is a reverse Carleson measure for the space $HB_0^{p,q}$ if and only if $\mu(I) \geq C|I|$ for every arc $I \subset \partial\mathbb{D}$ in a standard way: by the application of the functions k_λ^l with an integer l such that $pl > 1$.
>
> On the other hand, if $0 < q \leq p < 1$, then there are no reverse Carleson measures for the space $HB_0^{p,q}$: it can be deduced from the embedding $HB_0^{p,q} \subset HF_0^{p,q}$ (see [21, Theorem 4.1]) and the fact that in this case there are no reverse Carleson measures for the space $HF_0^{p,q}$ due to Theorem 5.1.
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> Hence only the following case remains open.
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> **Problem 8.1.** Is it true that if $0 < p < 1$ and $p < q < 2$ then there are no reverse Carleson measures for the space $HB_0^{p,q}$?

Figure 1: Problem 8.1 and its parameter context in [1, p. 22].

For $0 < a, q < \infty$, use normalized Lebesgue measure m on $\mathbb{T}$ and write

$$\mathrm{M}_a(r,g) = \left(\int_{\mathbb{T}} |g(r\zeta)|^a \, dm(\zeta) \right)^{1/a}.$$

At smoothness zero, equivalent quasinorms for the holomorphic Besov and Triebel–Lizorkin spaces are

$$\|f\|_{HB_0^{a,q}} = |f(0)| + \left(\int_0^1 \mathrm{M}_a(r,f')^q (1-r)^{q-1} \, dr \right)^{1/q}, \tag{1}$$

$$\|f\|_{HF_0^{a,q}} = |f(0)| + \left(\int_{\mathbb{T}} \left(\int_0^1 |f'(r\zeta)|^q (1-r)^{q-1} \, dr \right)^{a/q} dm(\zeta) \right)^{1/a}. \tag{2}$$

The source's reverse inequality for $HB_0^{p,q}$ has $a = p$ and right-hand exponent p.

2 Proof intuition

The threshold $q = 2$ is exposed by a normalized lacunary polynomial with N frequency blocks. Orthogonality, or the classical lacunary inequality, keeps every fixed Hardy norm bounded. On the other hand, each frequency has its own thin radial annulus on which its derivative dominates all other frequencies. Every annulus contributes order $N^{-q/2}$ to the qth power of either mixed analytic quasinorm, so their total size is $N^{1/q-1/2}$.

An arbitrary finite measure can concentrate anywhere in the closed disk, so a fixed polynomial need not have controlled $L^s(\mu)$ norm. Averaging all rotations removes this obstruction: the angular average at every point of the disk is a Hardy radial mean. Hence at least one rotation has bounded $L^s(\mu)$ norm, while the mixed analytic norms are rotation invariant. This contradicts any proposed reverse inequality when $q < 2$.

3 The theorem

Theorem 1. *Let $0 < a, s < \infty$ and $0 < q < 2$. There is no finite positive Borel measure μ on $\overline{\mathbb{D}}$, together with a constant C, such that either*

$$\|f\|_{HB_0^{a,q}} \le C\|f\|_{L^s(\mu)} \quad \textit{for every analytic polynomial } f, \tag{B}$$

or

$$\|f\|_{HF_0^{a,q}} \le C\|f\|_{L^s(\mu)} \quad \textit{for every analytic polynomial } f. \tag{F}$$

Consequently Problem 8.1 of [1] has an affirmative answer.

Lemma 1 (annular lacunary lower bound)**.** *There is an absolute integer $M > 2$ such that, for*

$$P_N(z) = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} z^{M^j},$$

one has, for every $0 < a < \infty$ and $0 < q < 2$,

$$\min\left\{ \|P_N\|_{HB_0^{a,q}}, \|P_N\|_{HF_0^{a,q}} \right\} \ge c_{a,q} N^{1/q-1/2}.$$

Proof. Take $M = 64$, put $n_j = M^j$, and consider the pairwise disjoint intervals

$$I_j = \left[1 - \frac{2}{n_j},\, 1 - \frac{1}{n_j}\right].$$

If $r \in I_j$, then

$$n_j r^{n_j - 1} \geq e^{-3} n_j.$$

Indeed, $r \geq 1 - 2/n_j$ and $(n_j - 1)\log(1 - 2/n_j) \geq -2(n_j - 1)/(n_j - 2) \geq -3$. The terms of lower degree satisfy

$$\sum_{k<j} n_k r^{n_k - 1} \leq \sum_{k<j} n_k < \frac{n_j}{M-1},$$

whereas $r \leq 1 - 1/n_j$ gives

$$\sum_{k>j} n_k r^{n_k - 1} \leq n_j e^{1/M} \sum_{\ell \geq 1} M^\ell e^{-M^\ell}.$$

For $M = 64$, the sum of the last two displayed upper bounds is strictly less than $e^{-3} n_j$. The reverse triangle inequality therefore yields a constant $c_0 > 0$, independent of j, N, r, such that

$$|P_N'(re^{it})| \geq \frac{c_0 n_j}{\sqrt{N}} \qquad (r \in I_j,\ t \in \mathbb{R}).$$

Using (3) in (1) and summing over the disjoint I_j gives

$$\begin{aligned}
\|P_N\|_{HB_0^{a,q}}^q &\geq \sum_{j=1}^{N} \left(\frac{c_0 n_j}{\sqrt{N}}\right)^q \int_{I_j} (1-r)^{q-1}\, dr \\
&= \frac{c_0^q (2^q - 1)}{q}\, N^{1-q/2}.
\end{aligned}$$

Because (3) holds for every angle, inserting it first in the inner radial integral in (2) gives the same lower bound for the $HF_0^{a,q}$ quasinorm. Taking qth roots proves the lemma. □

Proof of Theorem 1. If $0 < s \leq 2$, orthogonality and monotonicity of norms give

$$\|P_N\|_{H^s} \leq \|P_N\|_{H^2} = 1.$$

If $2 < s < \infty$, the frequency sequence (M^j) is Hadamard lacunary, so the classical Paley–Zygmund inequality for lacunary polynomials [2, Ch. V] gives

$$\|P_N\|_{H^s} \leq C_s \left(\sum_{j=1}^{N} \frac{1}{N}\right)^{1/2} = C_s.$$

Combining the two cases, with a suitable C_s one has $\|P_N\|_{H^s} \leq C_s$ for every finite $s > 0$. In particular, the exact range of Problem 8.1 uses only orthogonality.

For $0 \leq \theta < 2\pi$, set $P_{N,\theta}(z) = P_N(e^{i\theta} z)$. Fubini's theorem, followed by monotonicity of Hardy radial means, gives

$$\begin{aligned}
\frac{1}{2\pi} \int_0^{2\pi} \|P_{N,\theta}\|_{L^s(\mu)}^s\, d\theta &= \int_{\overline{\mathbb{D}}} \frac{1}{2\pi} \int_0^{2\pi} |P_N(e^{i\theta} z)|^s\, d\theta\, d\mu(z) \\
&\leq \mu(\overline{\mathbb{D}}) \|P_N\|_{H^s}^s \leq C_s^s \mu(\overline{\mathbb{D}}).
\end{aligned}$$

Thus some θ_N satisfies

$$\|P_{N,\theta_N}\|_{L^s(\mu)} \le C_s \mu(\overline{\mathbb{D}})^{1/s}.$$

Both mixed analytic quasinorms are rotation invariant. If (B) or (F) held, Lemma 1 and (3) would imply

$$c_{a,q} N^{1/q-1/2} \le C C_s \mu(\overline{\mathbb{D}})^{1/s}$$

for every N. This is impossible because $q < 2$. Taking $a = s = p$ with $0 < p < 1$ and $p < q < 2$ proves the assertion in Problem 8.1. $\square$

4 Scope, checks, and novelty

The theorem is stronger than the source problem in two directions: it covers both HB and HF, and permits an arbitrary finite right-hand exponent s. It supplies the complete negative fine-index range $q < 2$ of the source's more general second problem. It makes no assertion at $q = 2$, for $s = \infty$, or about the remaining positive regimes.

The proof uses only polynomials. The derivative estimate is pointwise, so no convexity is used when $a < 1$. Boundary mass and atoms in μ are harmless because the rotation identity integrates directly over the closed disk.

A bounded search on 13 August 2026 used the exact text of Problem 8.1, arXiv:2412.02354, and combinations of "reverse Carleson", "Besov", "$p < 1$", "$p < q < 2$", and "lacunary". The current arXiv version still states the problem as open, and no later explicit resolution or matching argument was found. Novelty confidence is moderate: the ingredients are classical, but the rotation-averaged combination appears not to have been recorded for this problem.

Human-review recommendation. Check the numerical separation in Lemma 1, the all-s form of the standard lacunary inequality, and the Fubini/rotation step. The exact $p < 1$ source problem uses only orthogonality in place of the lacunary theorem.

References

[1] E. Doubtsov, A. Tselishchev, and I. Vasilyev, *Reverse Carleson measures for spaces of analytic functions*, arXiv:2412.02354 (2024).

[2] A. Zygmund, *Trigonometric Series*, 2nd ed., Vols. I–II, Cambridge University Press, 1959.